
Speculative Decoding: What Lossless Means, What It Doesn't, and What's Next

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Abstract

Speculative decoding now runs under nearly every hosted LLM, and the field reports success in acceptance rate and token speed while claiming the result is lossless. This education material is organized around three questions: What Lossless Means (how does speculative decoding work, and why is it lossless in theory?), What It Doesn't (does lossless inference hold true beyond math and code?), and What's Next (who owns the quality of accelerated inference?). Learners walk through three generations of draft models: EAGLE-3, DFlash, DSpark. Token speed and acceptance rate get reported, mostly on math and code. What is missing is token quality, unmeasured everywhere else. A hands-on lab on a single GPU reproduces the numbers: serve an accelerated model, and grade the outputs. All materials, including the interactive article and lab scripts, are publicly available through our interactive website¹.

1 Concept Summary

The market wants more tokens. Agent workflows chain dozens of model calls per task, reasoning models spend thousands of tokens thinking before they answer, and every product built on either one pays for latency twice, in compute and in user patience. The ideal answer is lossless inference acceleration: serve the same model faster without changing what it outputs. That demand has pulled acceleration steadily deeper into the model itself. Draft models for speculative decoding evolved through three generations in two years: **EAGLE-3** reuses target hidden states for feature-level drafting (~ 3.9 tokens/pass; Li et al., 2025); **DFlash** drafts a whole block in one diffusion pass (~ 4.4 ; Chen et al., 2026); **DSpark** adds a sequential head and confidence-scheduled trimming (~ 5.1 , Qwen3-4B code average; DeepSeek, 2026, Table 1). Frontier labs moved in the same direction from the training side: DeepSeek pushed FP8 into pre-training (DeepSeek-AI, 2024), gpt-oss shipped MXFP4 with quantization-aware training (OpenAI, 2025), and Kimi K3 ships the speculator itself, fine-tuned and validated before release (Kimi Team, 2026). The interactive walkthrough carries one worked example across all four decoding stages; in the DSpark stage a low-confidence draft token is trimmed before verification and corrected by the target, which is where lossless ends.

Inference providers now run some combination of quantization, speculative decoding, and serving-level optimizations on nearly every hosted model. What no one owns is the quality of the accelerated output. *What's measured*: a small draft proposes K tokens and the target verifies all K in a single forward pass at nearly the cost of one, and rejection-sampling verification guarantees the output distribution exactly (Leviathan et al., 2023; Chen et al., 2023). Speed and acceptance are easy to measure and quality is not, so evaluation collapsed onto acceptance length and tokens/s; draft training converged onto the same number: the LK loss is the negative log of per-token acceptance probability (Samarin et al., 2026; Kimi Team, 2026, §4.1.4). *What's missed*: in the evaluation harness of DeepSpec, DeepSeek's open-source draft training and evaluation stack, the nine standard datasets supply only prompts, with no grader or accuracy metric anywhere in the code, so the lossless claim rests entirely on a theorem that covers exact verification only; quantization, KV routing,

¹Interactive Website: <https://neurips2026-speculative-decoding.vercel.app/>

disaggregation, and relaxed acceptance sit outside it, and the measured behavior gap between owner-trained and vendor-assembled acceleration reaches 5.6 points on long-tail domains (LosslessBench, 2026), while verification concentrates in low-entropy math and code, exactly where acceleration is least likely to misbehave. *What’s next:* every published draft objective optimizes alignment or acceptance (Zhou et al., 2024; Bachmann et al., 2025); none trains a draft to preserve the target’s behavior on the long tail.

2 Prerequisites and Technical Level

Level: introductory-to-intermediate; accessible to any engineer or student who knows that an LLM generates text one token at a time. **Prerequisites:** basic familiarity with transformer inference (a forward pass produces a next-token distribution); no RL, training, or GPU-kernel background required. The rejection-sampling losslessness argument is presented intuitively, with the accept/repair rule optional for readers who want the math.

3 Learning Objectives and Outcomes

After engaging with this resource, a learner can:

- explain *why* speculative decoding is possible (verifying K tokens $\approx$ the cost of one), and read *acceptance length* correctly: accepted proposals per verification step, not saved passes;
- walk through EAGLE-3, DFlash, and DSpark, name the bottleneck each stage removes, and explain why training objective and evaluation metric converged onto the same number;
- **demonstrate hands-on** that acceptance rate is not accuracy: sweep the confidence threshold in the official evaluation code, grade the outputs, and watch the two numbers move in opposite directions;
- articulate the domain mismatch (verification concentrates in math and code, usage concentrates elsewhere) and pose the open question: what would a draft-training objective that preserves long-tail behavior look like?

4 Grounding in NeurIPS Research (2022–2026)

The three generations the resource teaches are themselves recent papers at NeurIPS and its sister venues:

- **EAGLE-3** (NeurIPS 2025): feature-level drafting with multi-layer fusion and a training-time test, the current autoregressive-draft baseline.
- **DFlash** (ICML 2026): a block-diffusion drafter that generates the whole block in one forward pass, over $6\times$ lossless acceleration and up to $2.5\times$ over EAGLE-3.
- **DSpark** (DeepSeek, 2026): confidence-scheduled verification on the block-parallel backbone, serving DeepSeek-V4-Flash 60–85% faster in production.

The walkthrough places them in the NeurIPS speculative-decoding line they extend, SpecTr (NeurIPS 2023), Sequoia (NeurIPS 2024), and SpecExec (NeurIPS 2024), alongside the foundational papers (Leviathan et al., 2023; Chen et al., 2023), FP8 pre-training in DeepSeek-V3 (2024), MXFP4 QAT in gpt-oss (2025), Judge Decoding (ICLR 2025), and Kimi K3 shipping the speculator as part of post-training (2026).

5 Teaching Materials Summary

All materials are original and created for this track: (1) an **interactive self-contained HTML article** following the What Lossless Means / What It Doesn’t / What’s Next arc, with a walkthrough with adjustable draft length; (2) a **hands-on lab** (single GPU): serve Qwen3-8B with the three released draft checkpoints (EAGLE-3 vs DFlash vs DSpark), reproduce published acceptance lengths, race the lanes on frontend and creative briefs, and measure per-domain acceptance across the five LosslessBench domains.

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